

A Multi Agent AI Study Planner and Academic Coach

AGENTIC AI Section 003

Spring 2026

Individual Project Proposal

Problem Statement and Motivation

Many students today have a very busy school life. They take many classes at the same time. Each class has homework exams projects and readings. Students must decide what to study first how long to study and when to study. This is very hard to do every day.

Most students use simple tools like calendars or task lists. These tools only store dates and notes. They do not think or plan. They do not change when a student falls behind or gets sick or feels tired. This causes stress and poor grades.

This project will build an AI system that helps students plan their school work. The system will act like a smart study helper. It will make plans track progress and change plans when needed.

The target users are college students online learners and students who also have jobs. These users need help to manage time and reduce stress.

An agentic AI system is the best solution because school work changes all the time. A fixed program cannot adjust. An agentic system can think decide and act by itself.

Use Cases and User Scenarios

Scenario One Weekly Planning

A student uploads class files like a syllabus. The student wants a weekly study plan. The agents will read the files find all deadlines and make a plan for each day. The plan will show what subject to study and how long to study.

Scenario Two Exam Study

A student has an exam soon and wants help to prepare. The agents will find what topics are in the exam. They will split the topics into small parts. They will make a daily study plan.

Scenario Three Falling Behind

A student missed homework and feels stressed.

The agents will find what work is late. They will see what is most important. They will make a new plan that the student can follow.

Initial System Design

The system will use many AI agents. Each agent has one main job.

Planner Agent makes long term study plans.

Scheduler Agent makes daily and weekly schedules.

Content Agent decides what topics to study.

Progress Agent checks what work is done.

Critic Agent checks if the plan is realistic.

Interface Agent shows results to the student.

These agents work together. The planner makes a plan. The scheduler puts it into time blocks. The critic checks if it is too hard. If it is too hard the plan is changed.

The system will use tools like file upload for class files calendar tools web search and a local database to save progress.

Feasibility and Risks

The system may sometimes make mistakes. It may read a file wrong or make a plan that is too hard. The critic agent will help reduce this.

There are also limits like API use student privacy and computer power. Student data will be kept safe.

Timeline

Week one and two project design

Week three and four front end

Week five and six multi agent system

Week seven and eight testing and results

Week nine final report and video

